INSTRUCTIONS TO STUDENTS FOR RESEARCH REPORT PREPARATION

Students must follow the guidelines below while preparing their research reports. The report should demonstrate technical depth, research novelty, rigorous experimentation, and clear documentation.

1. Abstract and Introduction

1. **Abstract:** The abstract must contain 150–200 words and briefly describe the research problem, motivation, proposed methodology, key results, and major conclusions.
2. **Introduction Diagram:** Include a simple diagram illustrating the research problem, overall concept, or proposed solution.
3. **Research Contributions:** Clearly specify the original contributions of the work in the last paragraph of the Introduction. Contributions should be specific, clearly stated, and measurable wherever possible.

2. Literature Review

4. **Minimum Length:** The literature review must contain at least 1 page of details, supported by relevant references.
5. **Critical Analysis:** Discuss existing methods, their strengths, limitations, and research gaps. Avoid merely listing or summarizing research papers.
6. **Research Gap Identification:** Clearly identify the research gap and explain how the proposed work addresses the limitations of existing approaches.

3. Methodology and Proposed Architecture

7. **Detailed Architecture Diagram:** Include a comprehensive architecture diagram showing all modules, data flow, processing stages, and innovative components of the proposed system.
8. **Complete Methodology:** Provide a complete description of the proposed approach, including the system architecture, algorithms, mathematical formulations, and relevant equations.
9. **Module-wise Explanation:** Explain every module in the architecture, describing its functionality, working principle, and role in the overall system. Include relevant mathematical equations wherever necessary.
10. **Algorithms:** Present the important algorithms using clear and concise pseudocode. Avoid lengthy textual explanations when the procedure can be expressed effectively through an algorithm.

4. Implementation and Experimental Setup

11. **Implementation Details:** Specify the software frameworks, hardware configuration, important parameter settings, and implementation procedures necessary to reproduce the proposed work.
12. **Dataset Description:** Clearly describe the datasets used, including their sources, characteristics, sample sizes, and preprocessing procedures.
13. **Dataset Splitting:** Report the training, validation, and testing splits, specifying the number of samples or percentages assigned to each subset. Ensure that the splitting strategy is appropriate and avoids data leakage.

5. Results and Evaluation

14. **Training and Evaluation Plots:** Include relevant training plots, such as training and validation loss and accuracy curves, wherever applicable. Use appropriate graphs to demonstrate convergence and model performance.
15. **Evaluation Metrics:** Evaluate the proposed approach using metrics appropriate to the research problem. Define the metrics clearly and report the corresponding results.
16. **Comparison with State-of-the-Art (SOTA) Methods:** Compare the proposed approach with relevant state-of-the-art methods using comparable datasets, evaluation metrics, and experimental settings. Clearly highlight the improvements achieved.
17. **Ablation Study:** Conduct an ablation study to assess the contribution of individual modules, components, or design choices. Demonstrate how removing or modifying each component affects overall performance.

6. Conclusion, Future Work, and Limitations

18. **Conclusion:** Summarize the major findings and contributions of the work. Ensure that all claims are supported by the experimental results and evidence presented in the report.
19. **Future Work:** Identify meaningful future research directions based on the findings, unresolved challenges, and opportunities for improvement.
20. **Limitations:** Discuss the limitations of the proposed approach in a separate subsection. Specify relevant constraints, assumptions, computational requirements, dataset limitations, and scenarios in which the method may not perform effectively.

7. General Report Preparation Guidelines

21. **Organization:** Maintain a clear, logical, and consistent structure throughout the report.
22. **Figures and Tables:** Number and label all figures, tables, equations, and algorithms appropriately. Explain each important figure and table in the main text.
23. **References:** Cite relevant and recent research papers wherever appropriate, following a consistent referencing style.
24. **Original Contributions:** Clearly distinguish the novel contributions of the proposed work from existing methods.
25. **Accuracy and Reproducibility:** Ensure that the reported results are accurate, reproducible, and supported by experimental evidence.
26. **Writing Quality:** Avoid unsupported claims, redundant explanations, unnecessary repetition, and ambiguous statements.

Important

The report must demonstrate not only the implementation of the proposed system but also its technical novelty, scientific justification, experimental validation, and limitations. Every major design decision and claimed improvement must be supported by appropriate reasoning or evidence.

Deadline for submission : 01-November-2026 @ 23:59 PM.